

Toward a Dual-Zeta / Nilpotent Fiber View of Prime Spectral Structure

Version 3 — Revised and Extended

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Abstract

We extend the Riemann zeta function $\zeta(s)$ into the algebra of dual numbers $\mathbb{C}[j]/(j^2)$, where $j^2=0$ and $j \neq 0$, obtaining the tangent lift $\zeta_2(s)=\zeta(s)+j\eta\zeta'(s)$. We derive a dual Euler product showing that the nilpotent sector couples directly to prime structure. We prove that near a simple zero ρ the dual zeta admits the *infinitesimal lifting* $\zeta_2(s) \approx \zeta'(\rho)[(s-\rho)+j\eta]$. Under $s \mapsto 1-s$ the nilpotent part changes sign ($j \mapsto -j$), identifying $\text{Re}(s)=\frac{1}{2}$ as the unique nilpotent balance locus. We prove that the symmetric log-derivative field $\mathcal{P}(t) = -2\sum \Lambda(n)n^{-\frac{1}{2}} \cos(t \log n)$ on the critical line. We introduce a corrected symmetry-breaking functional $\Delta(s) := \text{Re}[(\zeta'/\zeta)(s) - (\zeta'/\zeta)(1-s)]$ and prove rigorously, via the Schwarz reflection principle, that $\Delta(s)=0$ on the critical line and $\Delta(\frac{1}{2}+\delta) = -\Delta(\frac{1}{2}-\delta)$ everywhere. Numerical computation over fifteen zeros (35-digit precision) confirms $\Delta=0$ on the critical line (residuals $< 10^{-10}$) and shows the energy ratio $\varepsilon(\frac{1}{3}, \gamma)/\varepsilon(\frac{1}{2}, \gamma)$ grows monotonically from $1.34\times$ to $2.23\times$. We state the *Nilpotent Symmetry Conjecture* (NSC): if $\zeta(\rho)=0$ and $\Delta(\rho)=0$ then $\text{Re}(\rho)=\frac{1}{2}$, which if proved implies RH. No proof of NSC is offered; this note is a geometric research programme.

Keywords:

Riemann zeta function, dual numbers, nilpotent algebra, infinitesimal lifting, tangent lift, prime flux field, von Mangoldt function, Hardy space, symmetry-breaking functional, fiber energy, Schwarz reflection principle,

functional equation, critical line, Riemann Hypothesis, odd Fourier
obstruction, explicit formula, prime shadow field, GUE statistics, Berry-
Keating operator, analytic number theory

1. Introduction

The nontrivial zeros of the Riemann zeta function govern the distribution of primes. The Riemann Hypothesis (RH) asserts $\text{Re}(\rho)=\frac{1}{2}$ for all such zeros. Two major geometric programmes inform our approach: Connes' noncommutative geometry [Con99], which realises zeros as an absorption spectrum on an adelic space; and the Berry–Keating programme [Ber86, Kea99], which seeks a self-adjoint operator whose spectrum reproduces zero heights. Deninger [Den98] proposed a cohomological framework with related goals.

We work in the *dual-number algebra* $\mathbb{C}[j]/(j^2)$, the algebraic foundation of automatic differentiation [Gri00] and synthetic differential geometry [Koc06], and organise the resulting tangent data as a geometric framework: nilpotent fibers, a helical path, and a symmetry-breaking functional. We do not claim dual numbers create new analytic content automatically; the novelty is *geometric organisation* of first-order jet data.

Throughout, claims are labelled:

- (T) Theorem [T]:** proved from standard identities.
- (H) Heuristic [H]:** geometric model, analytically motivated but unproved.
- (N) Numerical [N]:** computational observation at 35-digit precision.
- (S) Speculative [S]:** research direction for future work.

No claim is made toward a proof of RH. Version 2 corrects the definition and proof of the symmetry-breaking functional Δ (Section 8), extends numerical evidence to fifteen zeros, adds the nilpotent helix model (Section 7), and states NSC precisely.

2. Dual Number Extension [T]

Definition 2.1. The ring $\mathbb{C}[j]/(j^2)$ of *complex dual numbers* consists of elements $\alpha + j\beta$ ($\alpha, \beta \in \mathbb{C}$) with $j^2=0$, $j \neq 0$. It is isomorphic to the ring of first-order jets over $\mathbb{C}$, equivalently the tangent-bundle algebra [Koc06, §2.1]. For any analytic f and dual variable $\hat{s}=s+j\eta$:

$$f(s + j\eta) = f(s) + j\eta f'(s),$$

since all higher terms vanish by $j^2=0$. This is the standard identity underlying automatic differentiation [Gri00].

Definition 2.2. The *dual zeta function* is the tangent lift

$$\zeta_2(s) := \zeta(s + j\eta) = \zeta(s) + j\eta \zeta'(s),$$

for s in the region of convergence of $\zeta(s)$ and $\zeta'(s)$, $\eta \in \mathbb{R}$. Ordinary ζ is recovered at $\eta=0$.

3. Dual Euler Product [T]

Since $\exp(-j\eta \log p) = 1 - j\eta \log p$ and $(a+jb)^{-1} = a^{-1} - jba^{-2}$:

$$(1 - p^{-(s+j\eta)})^{-1} = (1-p^{-s})^{-1} - j\eta(\log p) \cdot p^{-s} (1-p^{-s})^{-2}.$$

Taking the product over all primes and applying $-\zeta'(s)/\zeta(s) = \sum_p (\log p)/(p^s - 1)$ [Tit86, §3.4] yields:

$$\zeta_2(s) = \zeta(s) \cdot (1 - j\eta \cdot \zeta'(s)/\zeta(s)).$$

Remark 3.1 [H]. We call $\sum_p (\log p)/(p^s - 1)$ the *prime tangent field*: the infinitesimal response of the Euler product to the shift $s \mapsto s + j\eta$. It is not a new analytic object; it is the logarithmic derivative of ζ organised as a nilpotent coefficient.

4. Infinitesimal Lifting of Simple Zeros [T]

Proposition 4.1. Let ρ be a simple zero of $\zeta(s)$. Then:

$$(i) \quad \zeta_2(\rho) = j\eta\zeta'(\rho) \neq 0.$$

$$(ii) \quad \text{Near } \rho : \zeta_2(s) \approx \zeta'(\rho) \cdot [(s-\rho) + j\eta].$$

Proof. (i) $\zeta(\rho)=0$ gives $\zeta_2(\rho)=j\eta\zeta'(\rho)$; simplicity of ρ gives $\zeta'(\rho)\neq 0$. (ii) $\zeta(s)\approx(s-\rho)\zeta'(\rho)$ and $\zeta'(s)\approx\zeta'(\rho)$ near ρ , so $\zeta_2(s)\approx\zeta'(\rho)[(s-\rho)+j\eta]$. $\square$

Remark 4.2 [H]. We call (ii) the *infinitesimal lifting* of ρ . The zero of ζ persists; the dual extension remembers the tangent direction at ρ , replacing the vanishing point by a non-vanishing first-order fiber $\{\rho+j\eta : \eta \in \mathbb{R}\}$ in $(\text{Re } s, \text{Im } s, j)$ -space.

Remark 4.3 [T] (Functional equation chirality). Under $s \mapsto 1-s$ applied to $\hat{s}=s+j\eta$: $1-\hat{s}=(1-s)-j\eta$. The nilpotent part changes sign: $j \mapsto -j$. The critical line $\text{Re}(s)=\frac{1}{2}$ is the unique fixed-point locus of $s \mapsto 1-s$ and hence the unique locus where the two nilpotent orientations $+j$ and $-j$ are equidistant from the axis.

5. Dual Lift of the Explicit Formula [H]

The following is formal; convergence of the lifted sum and branch behaviour of $\text{Li}(x')$ are not treated. It is a heuristic organising principle.

Lifting $\rho \mapsto \rho + j\eta$ in $J(x) = \text{Li}(x) - \sum_{\rho} \text{Li}(x^{\rho}) + \dots$:

$$\text{Li}(x^{\{\rho + j\eta\}}) = \text{Li}(x^{\rho}) + j\eta \cdot x^{\rho}/\rho.$$

Summing over zeros yields the *dual explicit formula* (formal):

$$J_2(\mathbf{x}) = J(\mathbf{x}) - j\eta \cdot \Phi(\mathbf{x}), \quad \Phi(\mathbf{x}) := \sum_{\rho} x^{\rho}/\rho.$$

$\Phi(x)$ is the classical oscillatory sum over zeros [Tit86, §14]. The nilpotent sector encodes it as the tangent correction to the zero contribution.

6. Symmetric Prime Flux Representation [T]

Definition 6.1. The *symmetric log-derivative field* is

$$\mathcal{F}(s) := (\zeta'/\zeta)(s) + (\zeta'/\zeta)(1-s).$$

Proposition 6.2 [T]. On $s=\frac{1}{2}+it$:

$$\mathcal{F}(t) = -2 \sum_n \Lambda(n) n^{-\frac{1}{2}} \cos(t \log n).$$

Proof. At $s=\frac{1}{2}+it$, $1-s=\frac{1}{2}-it=\bar{s}$. By the Schwarz reflection principle [Tit86, §2.1], $\zeta(\bar{s})=\overline{\zeta(s)}$, so $(\zeta'/\zeta)(\bar{s})=\overline{(\zeta'/\zeta)(s)}$. Hence $\mathcal{F}(t)=(\zeta'/\zeta)(s)+\overline{(\zeta'/\zeta)(s)}=2\Re[(\zeta'/\zeta)(s)]$. Inserting $\zeta'/\zeta=-\sum\Lambda(n)n^{-s}$ and taking real part at $s=\frac{1}{2}+it$ gives the cosine representation. $\square$

Remark 6.3 [H]. This Dirichlet-series symmetrisation is related to standard explicit-formula technology [Tit86, §14.2]; the novelty is its interpretation as the prime harmonic flux of the nilpotent balance condition $j \mapsto -j$. Each prime p contributes oscillations at frequency $\log p$.

7. Nilpotent Helix: A Geometric Model [H]

This section is geometric/heuristic. Objects are well-defined; no new analytic theorem about zeros is proved.

7.1 Three-Axis Space

Work in $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ with coordinates $(x, y, z) = (\text{Re}(s), \text{Im}(s), j\text{-component})$. The standard complex plane occupies $z=0$. The j -axis is bounded by walls at $z=\pm A$ (reflecting $j^2=0$). The critical line $\text{Re}(s)=\frac{1}{2}$ is the plane $x=\frac{1}{2}$.

7.2 The Helix Parametrisation

Fix $x=\frac{1}{2}$. Define the *nilpotent helix* by:

$$(x(t), y(t), z(t)) = \left(\frac{1}{2}, t, A \cdot \sin(\omega(t) \cdot t)\right), \quad \omega(t) \sim \log t.$$

Observation 7.1 [N]. The Hardy Z-function $Z(t) = e^{i\theta(t)} \zeta(\frac{1}{2} + it)$ is real and oscillates with angular frequency $\omega(t) \sim \log(t)$, consistent with the helix parametrisation above. The zero spacing (wavelength) $\Delta t_n \sim 2\pi / \log(\gamma_n)$ therefore decreases as γ_n grows — the helix winds more tightly at greater heights. Zeros are crossings of the helix with the plane $z=0$, and the spacing matches the prime number theorem $\pi(x) \sim x / \log x$.

7.3 Why $x = 1/2$ Only?

Geometric Claim 7.2 [H]. The critical line $x=\frac{1}{2}$ is the unique locus where the helix is symmetric under $j \mapsto -j$ (equidistant from both walls $\pm A$). For $x \neq \frac{1}{2}$ the helix axis is displaced, breaking $j \mapsto -j$ symmetry (Remark 4.3). Analytically this symmetry breaking is measured by $\Delta(s)$ defined in Section 8.

The phase-space lifting $q=Z(t)$, $p=\eta Z'(t)$ is analogous to Berry–Keating’s xp philosophy [Ber86]; $j \mapsto -j$ under $s \mapsto 1-s$ parallels time-reversal symmetry in spectral interpretations. These analogies are not developed into theorems here.

8. Symmetry-Breaking Functional and Nilpotent Symmetry Conjecture

8.1 Corrected Definition and Rigorous Proof [T]

Definition 8.1. (Symmetry-breaking functional.) Define

$$\Delta(s) := \operatorname{Re} [(\zeta'/\zeta)(s) - (\zeta'/\zeta)(1-s)].$$

Remark 8.2. This definition replaces the incorrect formula in Version 1. Earlier we wrote $\Delta(s) = (\zeta'/\zeta)(s) - \overline{(\zeta'/\zeta)(1-\bar{s})}$, which does not vanish on the critical line and whose proof was flawed. The corrected functional takes the real part of the difference of log-derivatives at s and its functional-equation reflection $1-s$.

Proposition 8.3 [T]. $\Delta(s) = 0$ for all s on the critical line $\operatorname{Re}(s) = \frac{1}{2}$.

Proof. Let $s = \frac{1}{2} + it$. Then $1-s = \frac{1}{2} - it = \bar{s}$ (complex conjugate). By the Schwarz reflection principle applied to ζ [Tit86, §2.1]:

$$\zeta(\bar{s}) = \overline{\zeta(s)} \quad \Rightarrow \quad (\zeta'/\zeta)(\bar{s}) = \overline{(\zeta'/\zeta)(s)}.$$

Since $1-s = \bar{s}$, we obtain:

$$(\zeta'/\zeta)(s) - (\zeta'/\zeta)(1-s) = f - \overline{f} = 2i \cdot \operatorname{Im}(f),$$

where $f := (\zeta'/\zeta)(s)$. Hence $\Delta(s) = \operatorname{Re}[2i \cdot \operatorname{Im}(f)] = 0$. $\square$

Proposition 8.4 [T] (Antisymmetry). For any $s = \frac{1}{2} + \delta + it$ with $\delta \in \mathbb{R}$:

$$\Delta(\tfrac{1}{2} + \delta + it) = -\Delta(\tfrac{1}{2} - \delta + it).$$

Proof. Let $s = \frac{1}{2} + \delta + it$ and $s' = \frac{1}{2} - \delta + it$. Then $1-s = \frac{1}{2} - \delta - it$ and $1-s' = \frac{1}{2} + \delta - it$. One computes:

$$\Delta(s) + \Delta(s') = \operatorname{Re}[(f(s) - f(1-s)) + (f(s') - f(1-s'))].$$

Applying the reflection principle $\zeta(\bar{s}) = \overline{\zeta(s)}$ and regrouping, the sum vanishes. $\square$

Remark 8.5 [N]. Numerical verification (35-digit precision, mpmath [Joh13]): $\Delta(\frac{1}{2} + it) = 0$ for all fifteen test zeros with $\operatorname{residual} < 10^{-10}$. At $\gamma_1 = 14.135$: $\Delta(0.1 + i\gamma_1) = -5.047$, $\Delta(0.4 + i\gamma_1) = -20.012$, $\Delta(0.6 + i\gamma_1) = +20.012$, $\Delta(0.9 + i\gamma_1) = +5.047$, confirming exact antisymmetry about $\sigma = \frac{1}{2}$.

8.2 Fiber Energy [T/N]

Definition 8.6. (Fiber energy.) For $\eta > 0$:

$$\varepsilon(s) := |\zeta(s)|^2 + \eta^2 |\zeta'(s)|^2.$$

Remark 8.7 [T]. At a simple zero ρ : $\zeta(\rho) = 0$ so $\varepsilon(\rho) = \eta^2 |\zeta'(\rho)|^2 > 0$. Energy is entirely carried by the nilpotent (derivative) sector, realising the infinitesimal lifting picture of Section 4.

Open Problem 8.8 [H]. Is the map $\sigma \mapsto \varepsilon(\sigma + i\gamma)$ strictly convex with unique global minimum at $\sigma = \frac{1}{2}$ for each γ corresponding to a nontrivial zero? Proving this would require computing $\partial \varepsilon / \partial \sigma = 2\operatorname{Re}[\zeta \zeta'] + 2\eta^2 \operatorname{Re}[\zeta' \zeta'']$ and establishing its sign change only at $\sigma = \frac{1}{2}$.

8.3 Numerical Evidence [N]

Table 1 reports $\varepsilon(\sigma, \gamma)$ at $\sigma \in \{\frac{1}{3}, \frac{1}{2}, \frac{2}{3}\}$ for the first fifteen nontrivial zeros, computed at 35-digit precision with $\eta=1$.

Table 1. *Fiber energy on and off the critical line.*

γ_n	$\varepsilon(\frac{1}{2}, \gamma)$	$\varepsilon(\frac{1}{3}, \gamma)$	$\varepsilon(\frac{2}{3}, \gamma)$	$\varepsilon(\frac{1}{3})/\varepsilon(\frac{1}{2})$	$\varepsilon(\frac{2}{3})/\varepsilon(\frac{1}{2})$
14.135	0.62910	0.84473	0.49571	1.343×	0.788×
21.022	1.29240	1.97913	0.89472	1.531×	0.692×
25.011	1.88162	3.04998	1.23000	1.621×	0.654×
30.425	1.70026	2.95654	1.04979	1.739×	0.617×
32.935	1.91025	3.41019	1.14959	1.785×	0.602×
37.586	3.75006	6.94152	2.14130	1.851×	0.571×
40.919	2.22192	4.26020	1.24477	1.917×	0.560×
43.327	3.36176	6.55749	1.84631	1.951×	0.549×
48.005	2.45872	5.01842	1.32682	2.041×	0.540×
49.774	2.01337	4.15999	1.07267	2.066×	0.533×
52.970	5.88829	12.18599	2.99897	2.070×	0.509×
56.446	5.60712	11.86240	2.80108	2.116×	0.500×
59.347	1.93712	4.27839	0.98603	2.209×	0.509×
60.832	2.73621	6.09773	1.38774	2.229×	0.507×
65.113	5.23393	11.70151	2.52633	2.236×	0.483×

Observation 8.9 [N]. Three patterns hold across all fifteen zeros: (i) the ratio $\varepsilon(\frac{1}{3}, \gamma)/\varepsilon(\frac{1}{2}, \gamma)$ increases *monotonically* from 1.34× to 2.24×; (ii) the ratio $\varepsilon(\frac{2}{3}, \gamma)/\varepsilon(\frac{1}{2}, \gamma)$ decreases from 0.79× to 0.48×; (iii) the product $[\varepsilon(\frac{1}{3}, \gamma)/\varepsilon(\frac{1}{2}, \gamma)] \times [\varepsilon(\frac{2}{3}, \gamma)/\varepsilon(\frac{1}{2}, \gamma)]$ stays near

1, consistent with a multiplicative antisymmetry. These patterns are consistent with the helix geometry but do not prove Open Problem 8.8.

8.4 The Conjecture [S]

Conjecture 8.10 (Nilpotent Symmetry Conjecture — NSC). *Let $\rho = \sigma + i\gamma$ be a nontrivial zero of $\zeta(s)$ with $0 < \sigma < 1$. Then:*

$$(\text{NSC}) \quad \Delta(\rho) = 0 \quad \Leftrightarrow \quad \sigma = 1/2.$$

Remark 8.11. The direction $\sigma = 1/2 \Rightarrow \Delta(\rho) = 0$ is Proposition 8.3. The substance of NSC is the converse: that $\zeta(\rho) = 0$ and $\Delta(\rho) = 0$ together force $\sigma = 1/2$. Note that $\Delta(s)$ can vanish off the critical line at points that are *not* zeros of ζ ; NSC constrains only the simultaneous condition. RH follows immediately from NSC: if all zeros satisfy $\Delta(\rho) = 0$ and NSC holds, then all zeros lie on $\sigma = 1/2$.

9. Speculative Operator Connections [S]

Everything here is heuristic. No rigorous spectral claims are made.

The fiber energy $\varepsilon = |\zeta|^2 + \eta^2 |\zeta'|^2$ suggests phase-space coordinates $q = \zeta$, $p = \eta \zeta'$. A formal dual operator Ansätze paralleling Berry–Keating [Ber86, Kea99]:

$$H_2 := -(\mathrm{d}/\mathrm{d}t + j\eta \cdot \mathrm{d}^2/\mathrm{d}t^2)^2 + V(t).$$

Since $j^2=0$: $H_2 = -\mathrm{d}^2/\mathrm{d}t^2 - 2j\eta \mathrm{d}^3/\mathrm{d}t^3 + V(t)$. Whether a potential V derived from $\Phi(x)$ could reproduce GUE zero-spacing statistics [BOPS] is entirely speculative. This Ansätze is recorded as a direction in the spirit of Deninger’s programme [Den98].

10. Future Directions

- (1) **Prove NSC.** Show $\zeta(\rho)=0 \wedge \Delta(\rho)=0$ implies $\sigma=1/2$. Possible approach: use known zero-free regions and the functional equation to bound Δ near putative off-line zeros.
- (2) **Energy monotonicity.** Prove or disprove: $\sigma \mapsto \varepsilon(\sigma, \gamma)$ has unique minimum at $\sigma=1/2$. Requires computing $\partial \varepsilon / \partial \sigma = 2\Re[\zeta \zeta'] + 2\eta^2 \Re[\{\zeta'\} \zeta'']$ and establishing its sign.
- (3) **Large-scale numerics.** Extend energy ratio to $\gamma > 10^3$. Does monotonicity persist? Regression of ratio vs $\log(\gamma)$ would clarify asymptotics.
- (4) **Function-field analogue.** Test NSC for zeta over $\square_u[C]$, where convergence issues are absent.
- (5) **Relation to Connes.** Determine if the nilpotent fiber picture is a shadow of the adelic spectral interpretation [Con99].
- (6) **Microlocal geometry.** The dual algebra is the ring of functions on the first-order jet bundle; connecting to microlocal analysis [Ö73] may rigourify the helix geometry.
- (7) **Connection to Islam's Hardy-space obstruction.** Islam [Isl26] identifies the precise RH obstruction as the odd Fourier component $f_{\text{e}} \square_t(t) = \text{Im}[-(\zeta'/\zeta)(1/2+it)]$, which is not constrained by the functional equation alone. Our nilpotent framework provides a geometric interpretation of this same structure: at a simple zero ρ , the dual zeta satisfies $\zeta_2(\rho) = j\eta\zeta'(\rho)$, and the derivative $\zeta'(\rho) = \text{Re}[\zeta'(\rho)] + i \cdot \text{Im}[\zeta'(\rho)]$ carries $\text{Im}[\zeta'(\rho)]$ — which corresponds to the odd component identified as the unconstrained obstruction. Whether fiber energy constraints on $\varepsilon(\rho) = \eta^2 |\zeta'(\rho)|^2$ can provide an independent bound on $\text{Im}[\zeta'(\rho)]$ is an open question. A positive result would connect the nilpotent framework directly to the millennium-prize obstruction of [Isl26].

11. Conclusion

Proved results [T]: dual Euler product (Section 3); infinitesimal lifting of simple zeros, Proposition 4.1; functional-equation chirality $j \mapsto -j$, Remark 4.3; prime harmonic representation $\mathcal{H}(t)$, Proposition 6.2; $\Delta=0$ on critical line via Schwarz reflection, Proposition 8.3; antisymmetry $\Delta(\frac{1}{2}+\delta)=-\Delta(\frac{1}{2}-\delta)$, Proposition 8.4.

Geometric contributions [H]: nilpotent helix model (Section 7); Nilpotent Symmetry Conjecture NSC (Conjecture 8.10).

Numerical evidence [N]: over fifteen zeros, $\Delta=0$ on critical line (residual $< 10^{-10}$); energy ratio $\varepsilon(\frac{1}{3}, \gamma)/\varepsilon(\frac{1}{2}, \gamma)$ monotonically grows $1.34\times \rightarrow 2.24\times$; multiplicative antisymmetry of ratios observed.

Key correction from V1: $\Delta(s)$ redefined as $\operatorname{Re}[(\zeta'/\zeta)(s) - (\zeta'/\zeta)(1-s)]$; the earlier conjugate-based definition was incorrect and its “proof” flawed. The corrected version has a rigorous proof via the Schwarz reflection principle.

Relation to Islam (2026): Islam [Isl26] identifies the precise RH obstruction as the odd Fourier component $\operatorname{Im}[-(\zeta'/\zeta)(\frac{1}{2}+it)]$, unconstrained by the functional equation. Our nilpotent framework provides a geometric interpretation: the fiber $\zeta_2(\rho) = j\eta\zeta'(\rho)$ encodes exactly this derivative, whose imaginary part is Islam’s unconstrained component. Whether the fiber energy $\varepsilon(\rho) = \eta^2|\zeta'(\rho)|^2$ can constrain $\operatorname{Im}[\zeta'(\rho)]$ independently remains an open problem connecting the two frameworks.

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